

Exp No: 25

PRODUCER-CONSUMER PROBLEM USING SYSTEMC

```
anish@DESKTOP-BUFCRBD: ~$ nano producer_consumer.cpp
GNU nano 8.7.1 producer_consumer.cpp
#include <systemc.h>

// Producer Module
SC_MODULE(Producer) {
    sc_fifo_out<int> out;

    void produce() {
        for (int i = 1; i <= 5; i++) {
            cout << "Produced: " << i << endl;
            out.write(i);
            wait(1, SC_SEC);
        }
    }

    SC_CTOR(Producer) {
        SC_THREAD(produce);
    }
};

// Consumer Module
SC_MODULE(Consumer) {
    sc_fifo_in<int> in;

    void consume() {
        int value;
        while (true) {
            in.read(value);
            cout << "Consumed: " << value << endl;
            wait(1, SC_SEC);
        }
    }
};
```

```
anish@DESKTOP-BUFCRBD: ~$ nano producer_consumer.cpp
anish@DESKTOP-BUFCRBD: ~$ g++ -o producer_consumer producer_consumer.cpp -lsystemc
anish@DESKTOP-BUFCRBD: ~$ ./producer_consumer

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED

Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
anish@DESKTOP-BUFCRBD: ~$
```